



EGEHAN YAĞLICI
GEOSPATIAL PORTFOLIO

REPORT 03 / 04

PROJECT REPORT / 3D CITY MODELLING

Integrated 3D City Modelling from LiDAR & Building Footprints

ArcGIS Pro, TerraScan and CloudCompare Workflow

A quality-controlled pipeline for footprint correction, height attribution, LoD2 roof modelling and point-to-model assessment.



ArcGIS Pro · LiDAR · TerraScan · TerraSolid · CloudCompare

egehanyaglici.com / REPORTS

2023

First, we clarify the aim of the project and the overall workflow. We work with two primary datasets: a building footprint shapefile and a LiDAR point cloud. Using these data, our objective is to identify and correct discrepancies between the point cloud and the shapefile that arise during processing and to derive improved 3D building information. After refining the consistency between footprints and heights in **ArcGIS Pro**, we inspect the roofs of the buildings in **MicroStation**, correct any detected errors, and finally assess the overall quality of the 3D model in **CloudCompare**.

When the existing data are transferred into ArcGIS Pro, the initial view of the point cloud is automatically displayed on top of the shapefile.



point cloud



shape file

We generated the point cloud data, stored it in a separate cloud file and then worked with it. We have classified the newly created cloud.las file. In this way, the system automatically separated the buildings, water areas, etc. from each other. We used the coordinate system coded 2180 in every operation we did.

•	0	0 Never Cla...
•	2	2 Ground
•	3	3 Low Vege...
•	4	4 Medium V...
•	5	5 High Veg...
•	6	6 Building
•	7	7 Noise
•	9	9 Water
•	12	12 Overlap...

Classification



Just Buildings



Ground

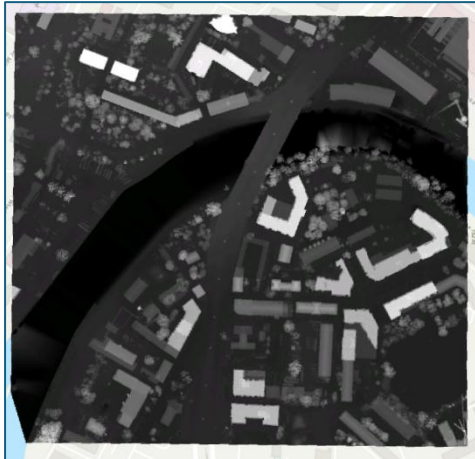


1st Return

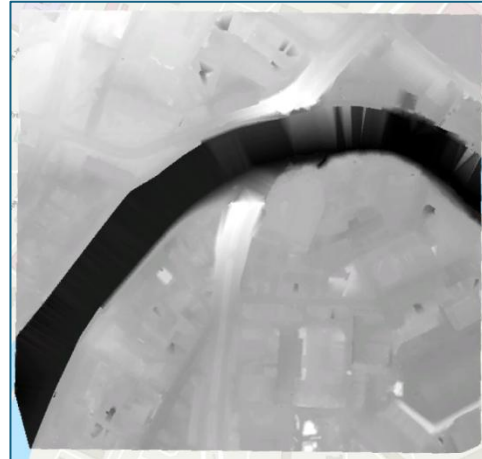


Low Vege.

After that, we produced the digital terrain model and digital surface model and we removed the DTM model from DSM.

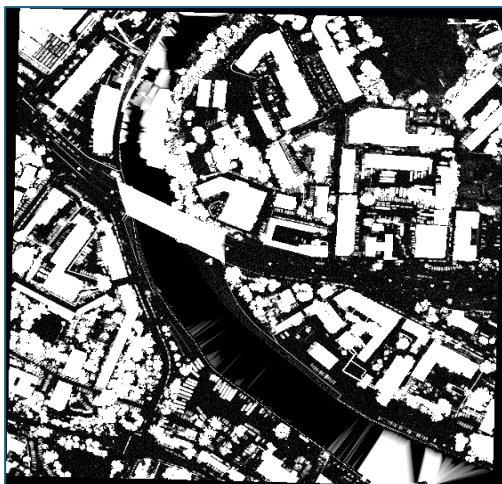


Digital Surface Model (DSM)



Digital Terrain Model (DTM)

We extracted a digital terrain model from the digital surface model ($DSM - DTM$).



After obtaining nDSM, we analyzed the data in our Cloud.lasd file from the Zonal Statistics as Table (Spatial Analyst Tool) section into our shape file with nDSM and transferred the resulting data as a table in shape file. We were able to create a 3D representation of the structures in our shape file using the information we collected as a result of this process.

nDSM

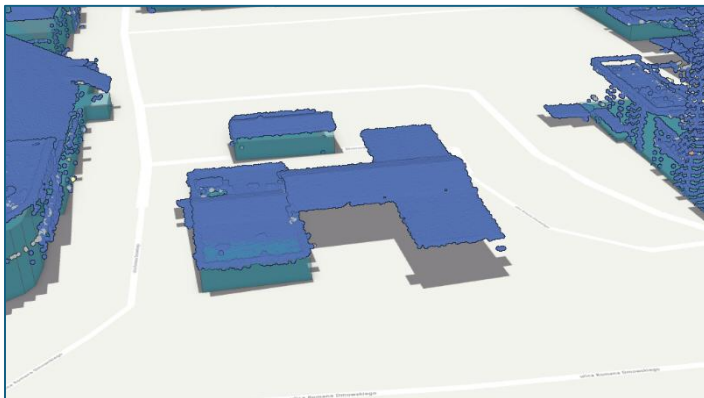


By importing our cloud.lasd file from the LAS Point Statistics As Raster Tool, we were able to create a file named bud.tif that contained only the building footprints and the most frequent class code method. In the continuation of our operations, we converted the Classification method in the Point Cloud to return number from the symbology part.

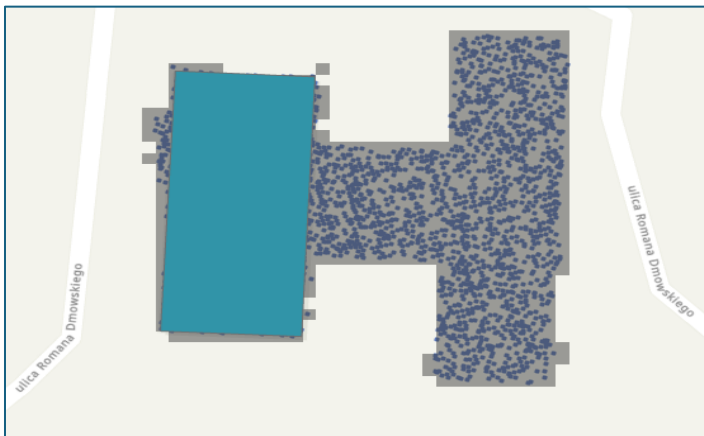


Now we reach the stage where we detect and correct errors in the building dataset. When we compare the point cloud with the 2D building layer, we observe that in some locations there are clear building points in the point cloud, while no corresponding polygon or height value exists in the shapefile. Using the raster footprint generated from the point cloud, we identify these gaps, locate the buildings that are missing in the 2D data, and manually digitize their footprints so that the geometric outlines of the buildings are properly represented. At this point, the building polygons exist in the 2D layer, but they still have no height attributes.

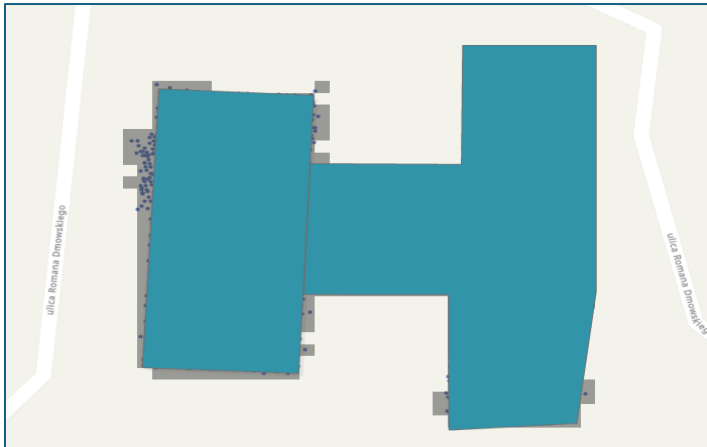
For the height value of this building, from the Zonal Statistics as Table (Spatial Analyst Tool) section. After choosing our shape file as the input raster, we choose our input value raster as the nDSM we found before, as we said at the beginning of the report, we perform all our operations with the 2180 coded coordinate system. With this process, we remove the table in the shape file from the join section and transfer our new table containing the corrected version. Instead of doing this one by one, I did it all at once after fixing the errors.



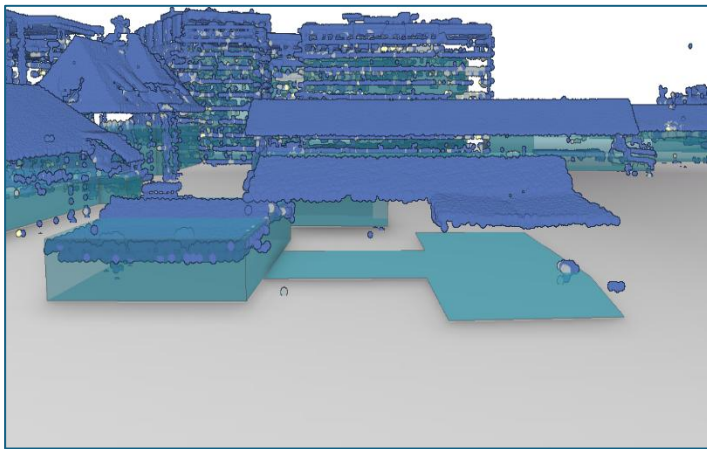
As you can see, while we have point cloud data, the building has no polygon and height.



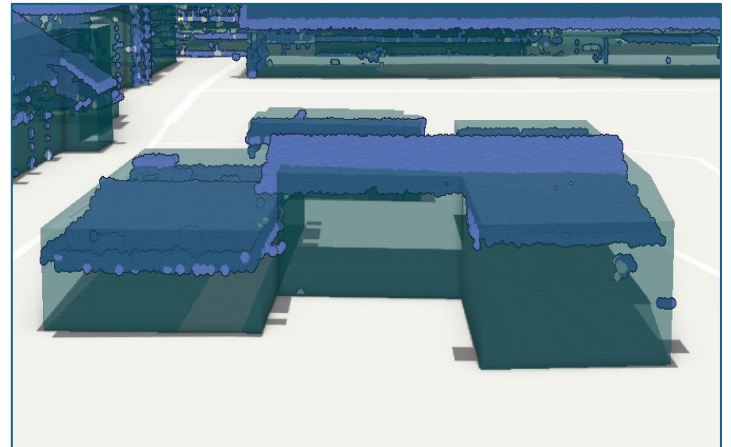
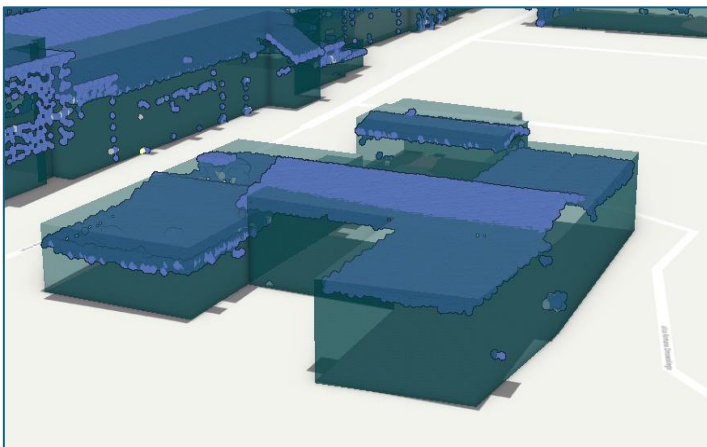
In the 2D view, we see that the polygon is not drawn, but the point cloud data exists. Here, we draw and complete this missing part by selecting Create polygon from the



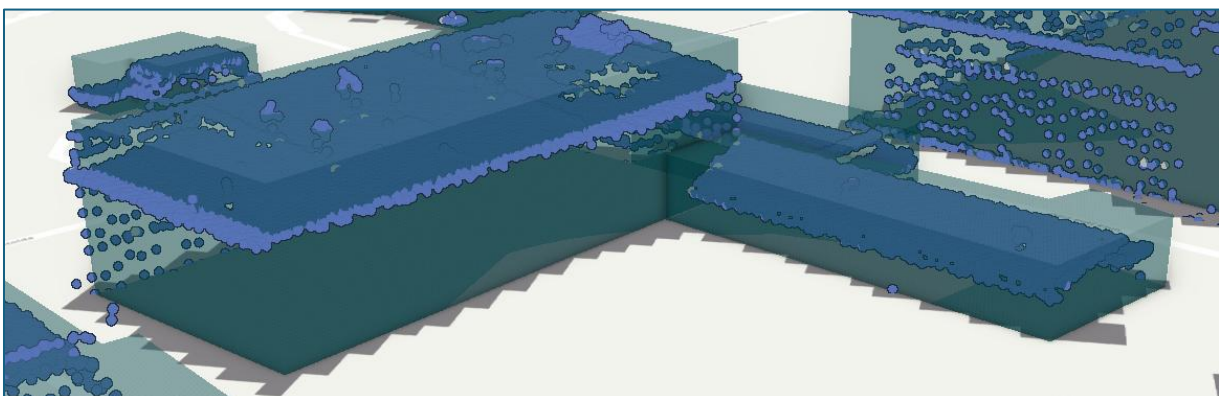
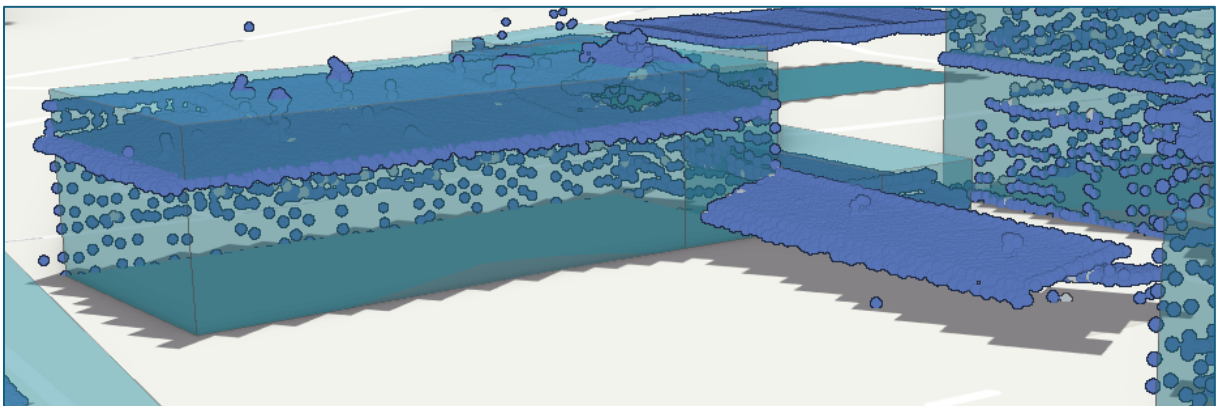
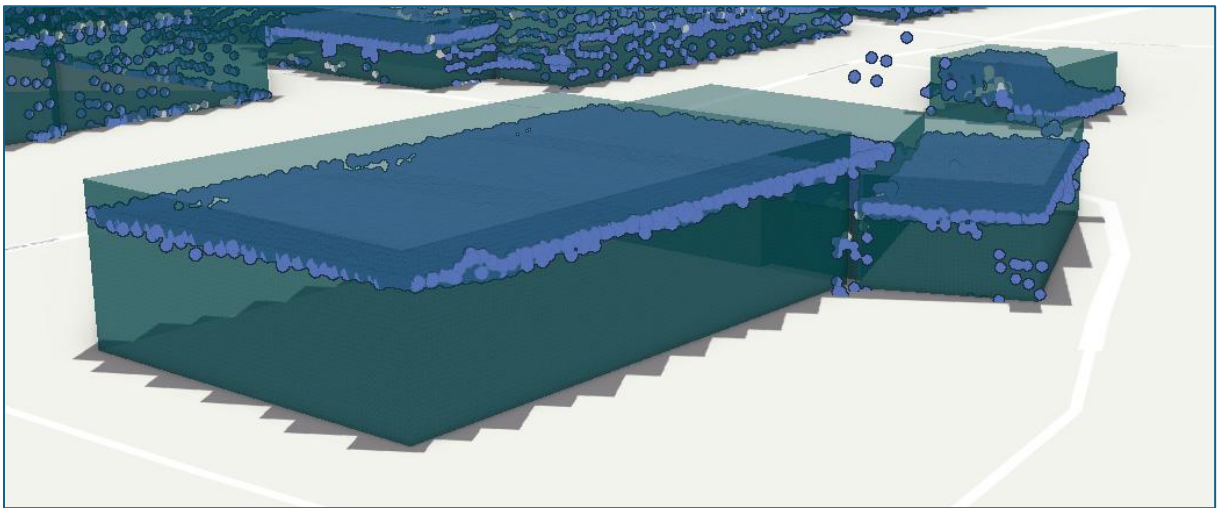
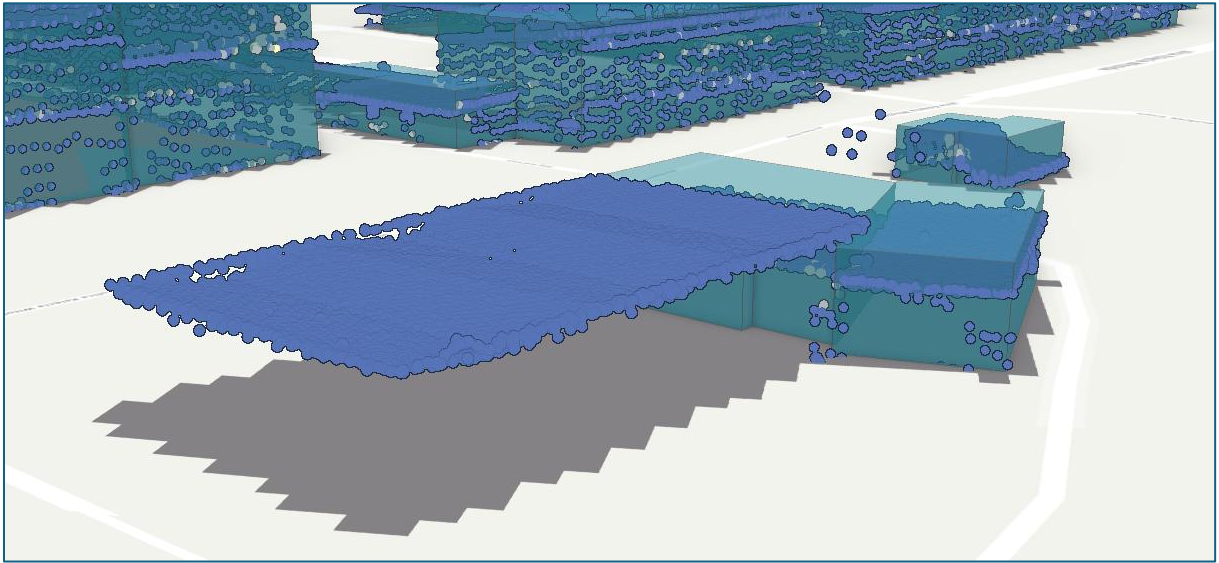
2D state after polygon drawing

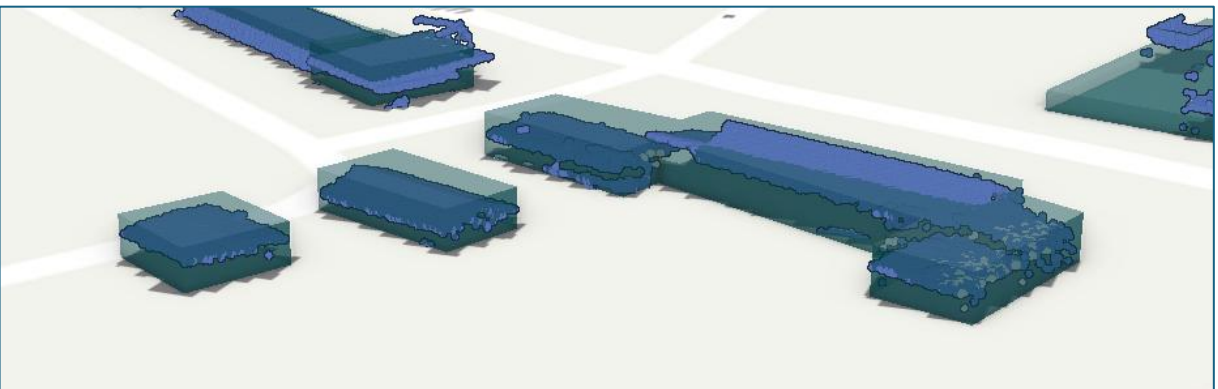
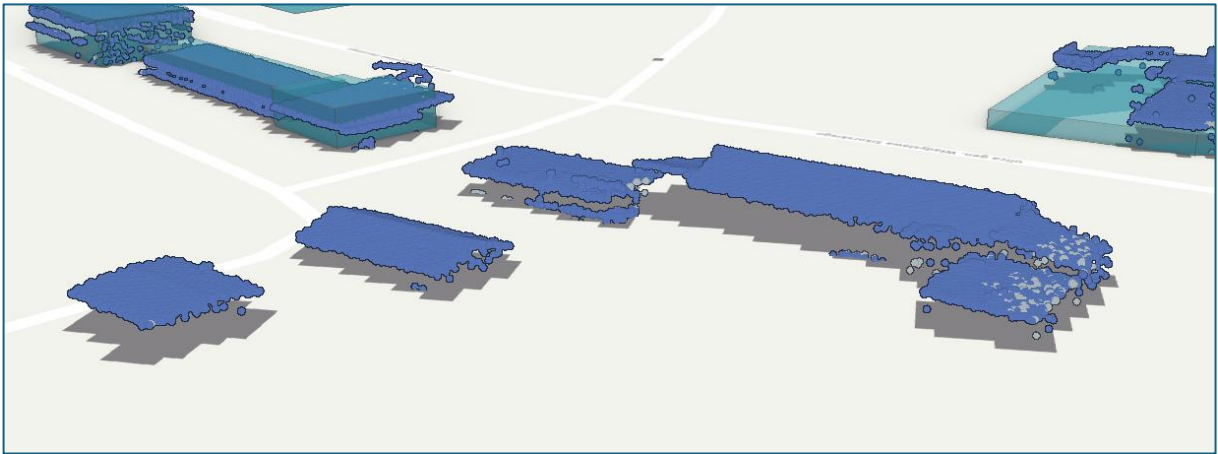
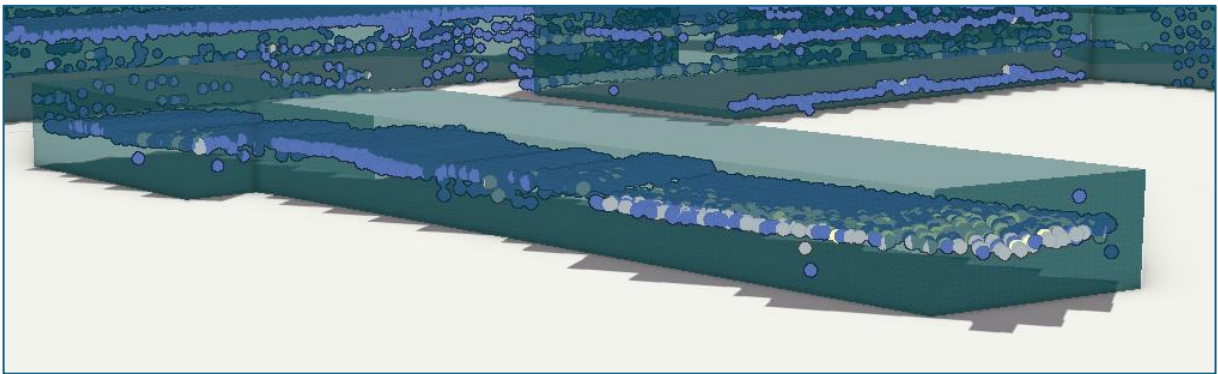
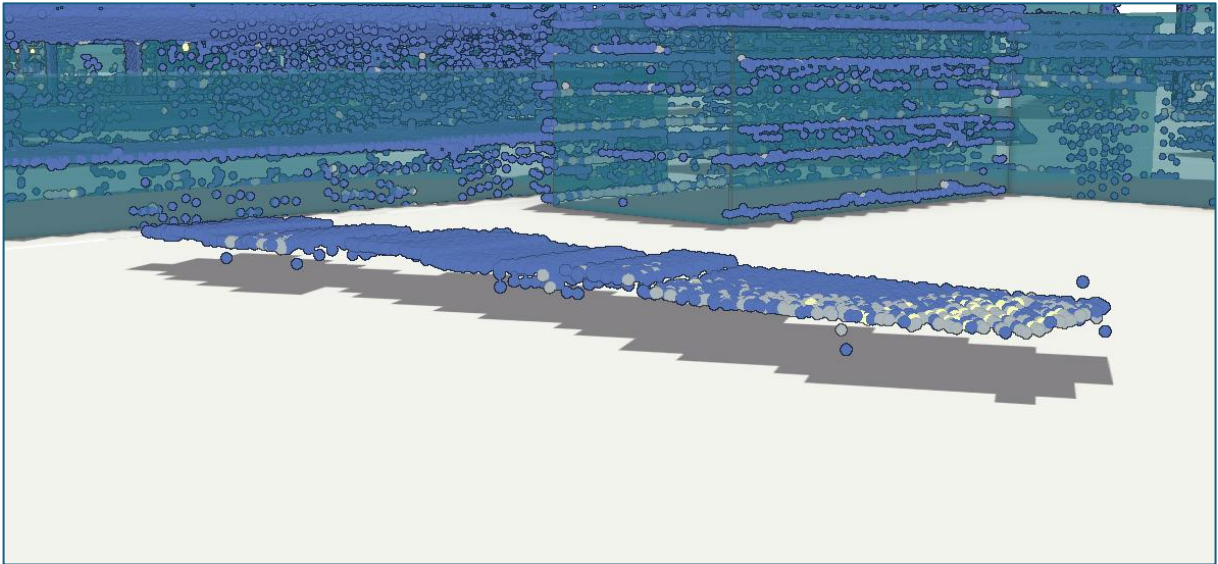


In 3D it looks like this. In this case, our data doesn't just have a height state. As we mentioned above, when we perform our operation with the Zonal Statistics as Table (Spatial Analyst Tool) and add the new table to our shape file, we see the new version with the height calculated below.



To exemplify a little more;





2D BEFORE



2D AFTER



Let's take a step-by-step look at the **operations** we do in the MicroStation program
(*We used TerraScan and TerraSolid tools in MicroStation*):

TerraScan is used to analyse LIDAR data and create 3D surface models. It has several different tools and capabilities for precisely classifying and arranging LIDAR data, including surface construction, point cloud filtering, categorization, editing, and image processing.

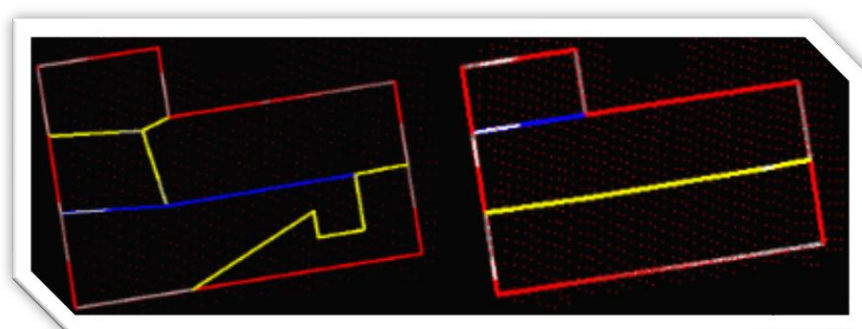
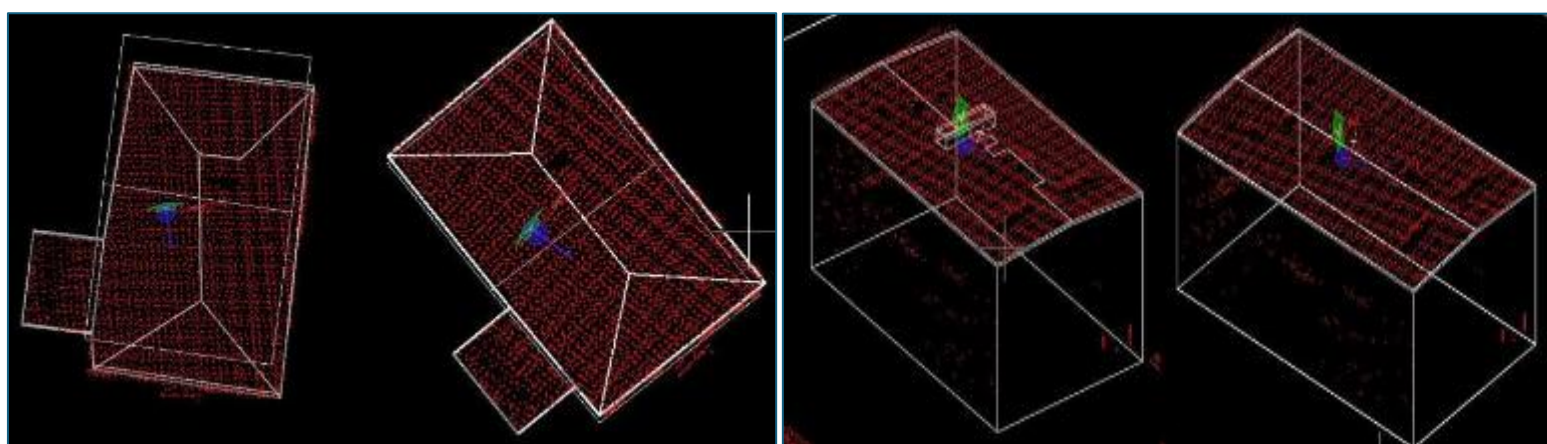
Another program for analyzing LIDAR data and creating 3D surface models is **TerraSolid**. It offers numerous tools and capabilities for processing, categorizing, organizing, analyzing, and visualizing LIDAR data, including point cloud categorization, surface construction, 3D visualization, land use analysis, and other comparable functions.

We transferred our shape file to MicroStation for Building Footprints

To categorize the point cloud data into different groups, such as ground points, vegetation points and building points we use the "Classification" tool. You may split and manipulate various types of points using this to your advantage. We only chose buildings by classification because our project requires them.

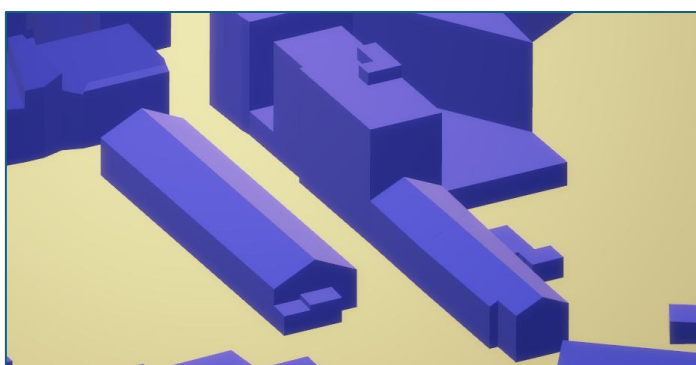
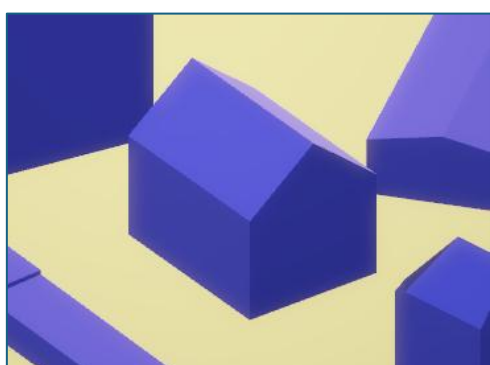
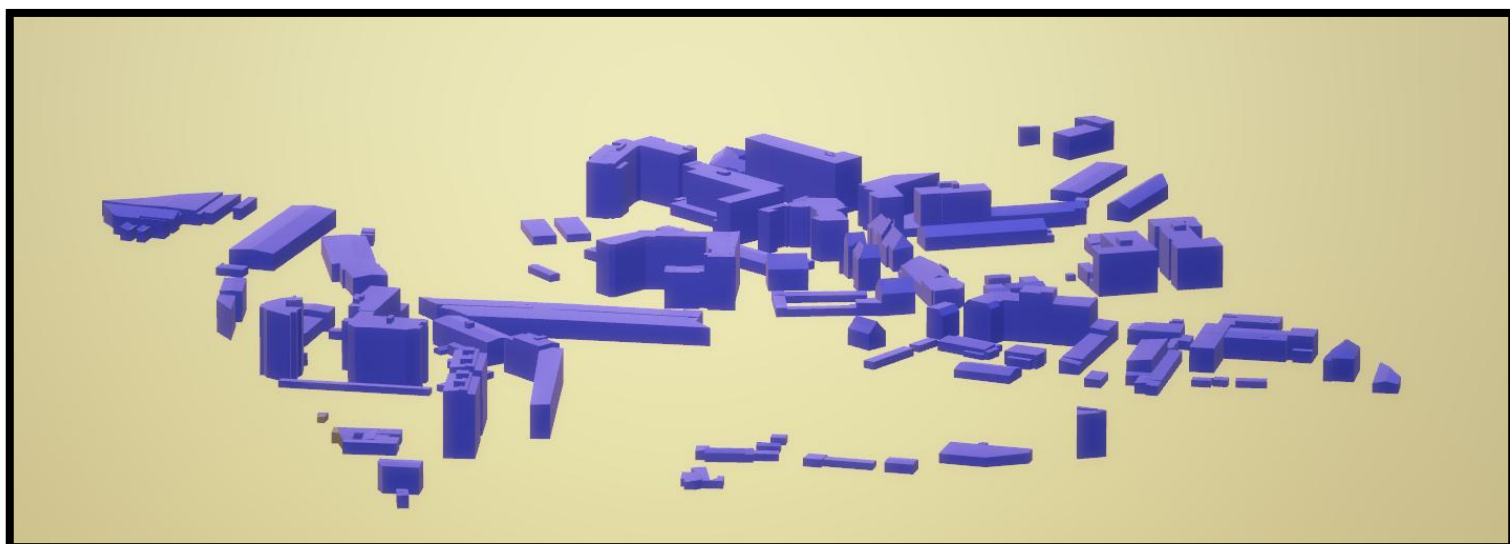
We continued by vectorizing the buildings we have. An essential step in verifying the correctness and completeness of the building data was vectorizing building roofs in the available data. The roof shapes were vectorized into formats that could be easily processed and studied using sophisticated software techniques. The procedure entailed locating the roof edges, correcting any mistakes or inconsistencies and turning them into exact vector shapes that faithfully reproduced the building roofs.

Overall, vectorizing building roofs in the available data was a valuable exercise that allowed us to improve the accuracy and completeness of the building data. The resulting building map and improved data quality have numerous applications in a range of fields, it can contribute to **better planning, design, and decision-making processes**.

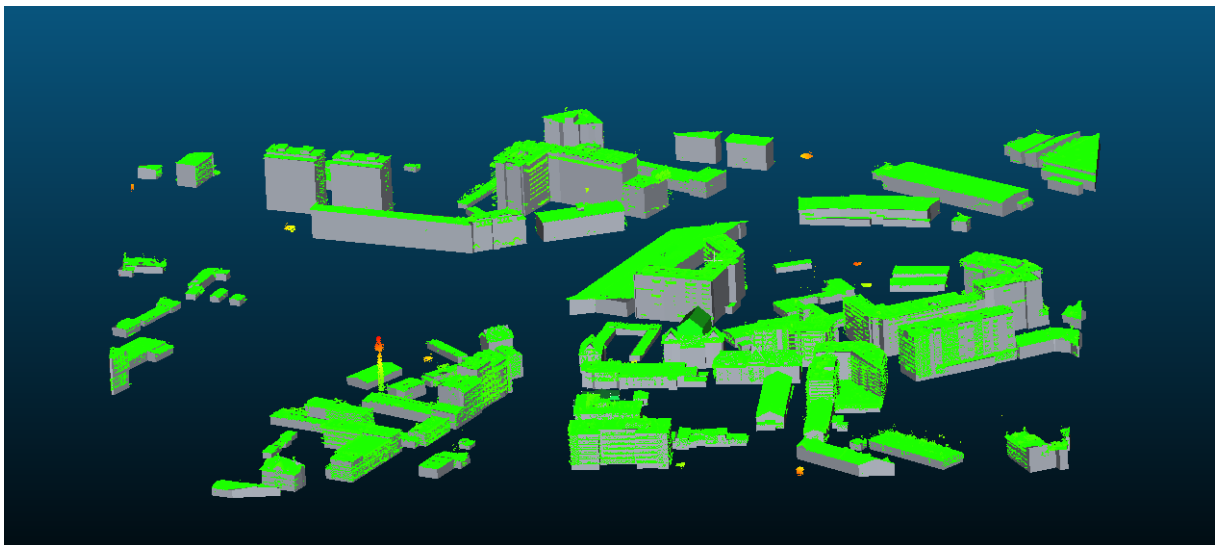
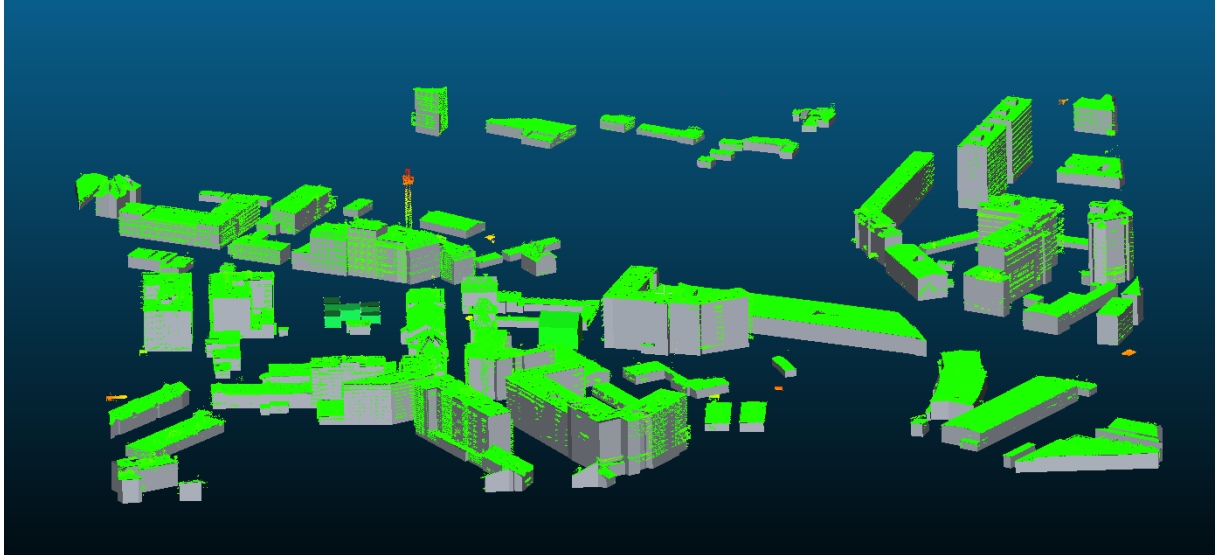


As you can see, the geometric defect on the building roofs has been fixed. After this stage, we need to get a 3D model of the data we have. For this, we use the export part of TerraSolid software.

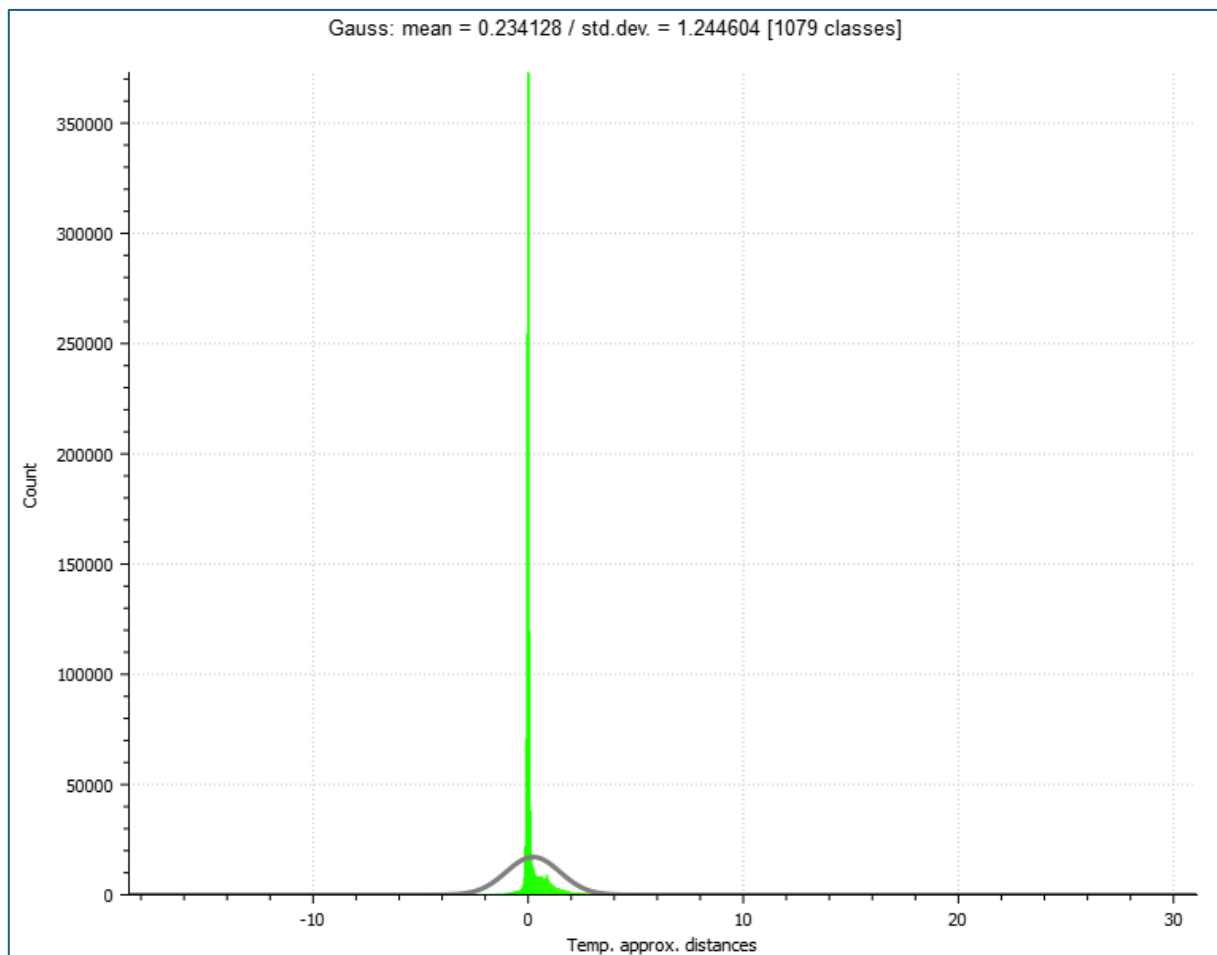
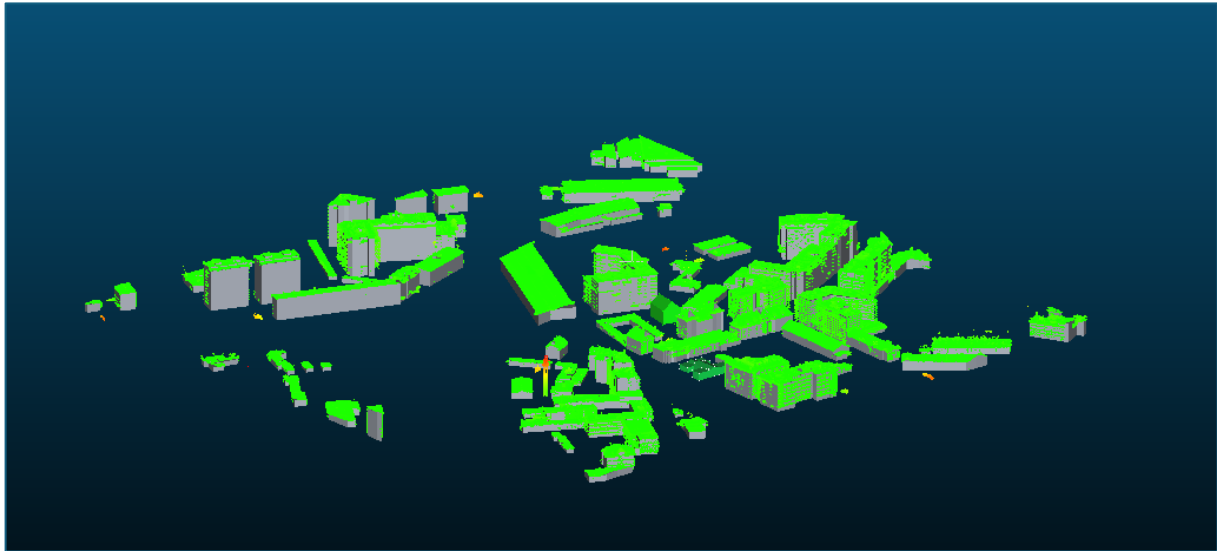
Let's examine the 3D model that we extracted and see how the roofs hold up in this model.

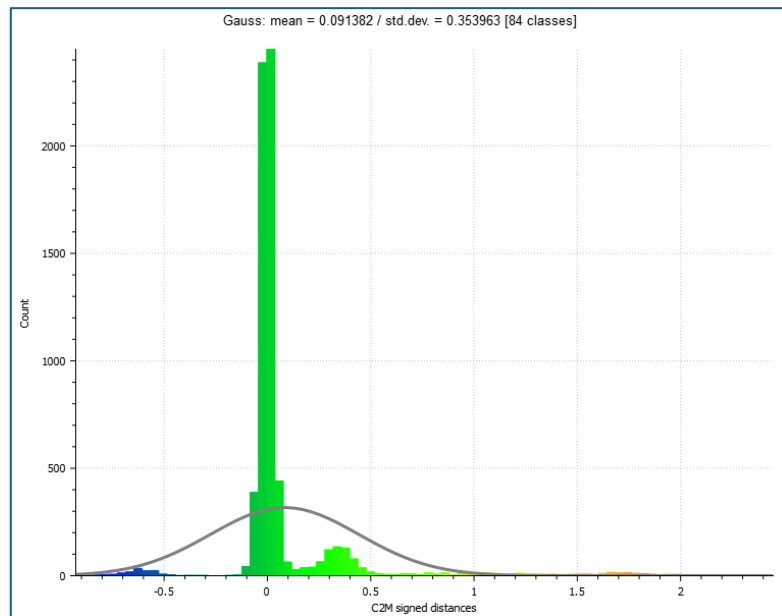
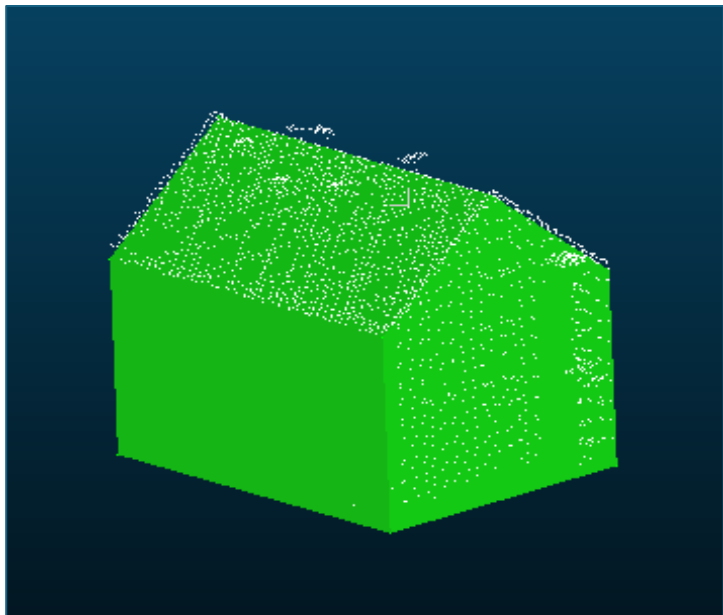
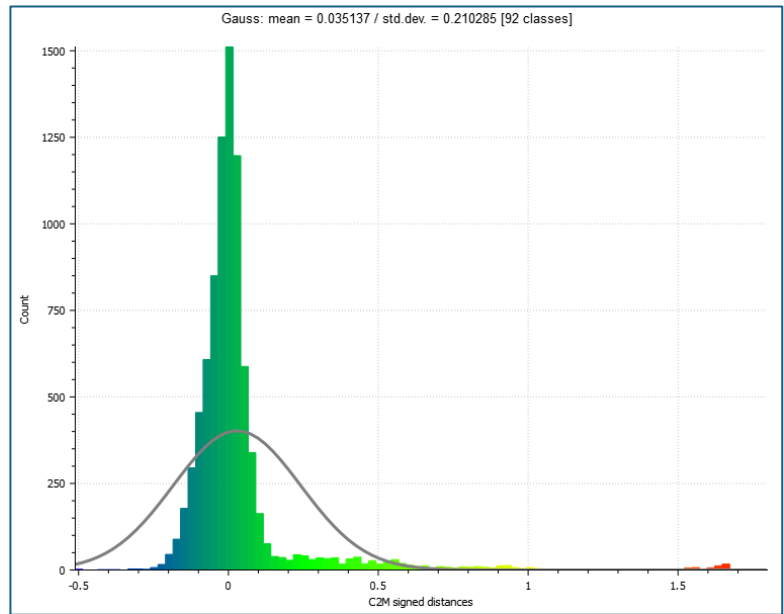
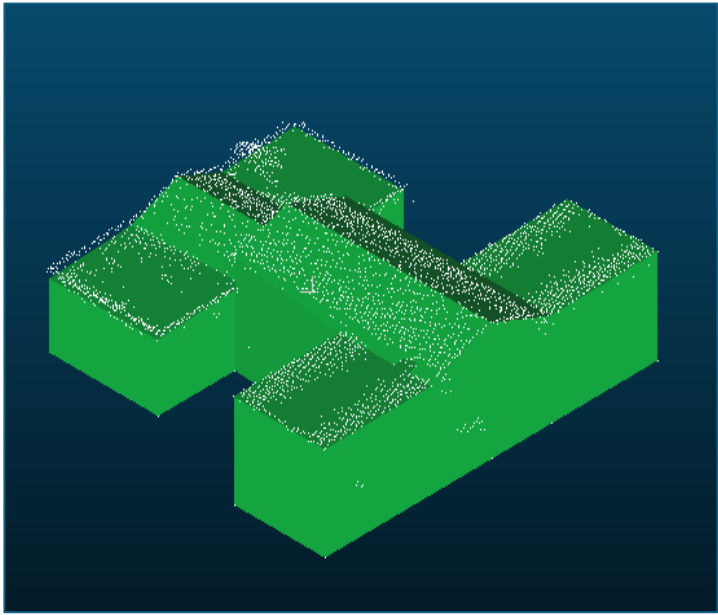


The final step involves transferring the 3D model we created and our point cloud data to the application, as well as using **CloudCompare** to evaluate the project's quality. We classify our point cloud data into **buildings** using the **classification** approach after loading the model and point cloud data into CloudCompare.



We utilize the 3D model we have and the point cloud data we have organized according to our intended purpose to compute the distance between the model and the point Cloud. After doing "**Distance Computation**" between our 3D model and our point cloud, we cut samples from our buildings and examine them separately with the **gaussian** method.





CONCLUSION

This project delivered a complete, end-to-end workflow for transforming raw LiDAR point clouds and existing building footprints into a consistent, quality-controlled 3D city model. Starting from a simple combination of a shapefile and a LAS dataset, we systematically aligned, enriched and refined the data until it reached a level of detail and reliability that can support real analytical and visualization tasks, not just classroom demonstrations.

On the ArcGIS Pro side, the building stock was first anchored on a solid geometric foundation. The point cloud was organized and classified, Digital Surface Model (DSM) and Digital Terrain Model (DTM) rasters were generated, and a normalized DSM (nDSM) was computed to express object heights above ground. Using zonal statistics, these heights were transferred to the building footprints so that each polygon carries meaningful 3D information rather than being a flat 2D outline. The additional use of a rasterized building footprint (derived from the point cloud classes) allowed us to detect and correct missing or incomplete buildings. Through targeted manual editing, we closed these gaps and ensured that the building layer matches what is actually present in the LiDAR data, not just what was digitized in the past.

In the second phase, MicroStation together with TerraScan and TerraSolid was used to push the models beyond simple extruded blocks. By isolating building points, vectorizing roof structures and correcting geometric defects, we produced Level of Detail 2 (LoD2) building models that reflect the real roof shapes instead of generic flat tops. This step is crucial for applications that depend on more realistic geometry, such as line-of-sight analyses, solar potential assessments or high-quality 3D visualizations. The result is a building dataset that is not only vertically accurate but also geometrically rich at the roof level.

Finally, the workflow was closed with a rigorous quality assessment in CloudCompare. By computing distances between the LiDAR point cloud and the exported 3D building surfaces, we obtained an objective measure of how well the model fits the measurements. Distance distributions and sample buildings were examined in detail, allowing us to distinguish random noise from systematic deviations. This step turns the project from a “we modelled something that looks nice” exercise into a traceable and defensible 3D data production process with quantifiable accuracy.

Taken together, these stages provide the client with much more than just a 3D file. They provide a documented, reproducible methodology for going from raw point clouds to usable 3D city models: a clear sequence of tools, parameters and checks that can be repeated on other areas or extended to larger regions. The resulting dataset, in a consistent coordinate system and with corrected footprints, meaningful height attributes and LoD2 roof geometries, is ready to be integrated into urban planning workflows, visualization platforms or further analytical pipelines. Future work can build directly on this foundation by automating more of the detection steps, adding textures or additional city objects, and scaling the process to full-city coverage, but the essential backbone of a robust 3D city modelling pipeline is now in place.